

Structures of LeuT in bicelles define conformation and substrate binding in a membrane-like context

Hui Wang¹, Johannes Elferich² & Eric Gouaux^{1,3}

Neurotransmitter sodium symporters (NSSs) catalyze the uptake of neurotransmitters into cells, terminating neurotransmission at chemical synapses. Consistent with the role of NSSs in the central nervous system, they are implicated in multiple diseases and disorders. LeuT, from *Aquifex aeolicus*, is a prokaryotic ortholog of the NSS family and has contributed to our understanding of the structure, mechanism and pharmacology of NSSs. At present, however, the functional state of LeuT in crystals grown in the presence of *n*-octyl- β -D-glucopyranoside (β -OG) and the number of substrate binding sites are controversial issues. Here we present crystal structures of LeuT grown in DMPC-CHAPSO bicelles and demonstrate that the conformations of LeuT–substrate complexes in lipid bicelles and in β -OG detergent micelles are nearly identical. Furthermore, using crystals grown in bicelles and the substrate leucine or the substrate analog selenomethionine, we find only a single substrate molecule in the primary binding site.

Neurotransmitter sodium symporters remove synaptically released glycine, γ -aminobutyric acid, serotonin, norepinephrine and dopamine to terminate signal transmission in the central nervous system by coupling neurotransmitter uptake to the sodium electrochemical gradient^{1–3}. The dysfunction of NSSs has been implicated in multiple central and peripheral nervous system diseases and disorders, including depression⁴, epilepsy⁵, orthostatic intolerance⁶, anxiety⁴ and Parkinson's disease⁴. NSSs are also primary targets of many therapeutic agents and addictive substances that include antidepressants⁷, antiepileptic medications⁸, anticonvulsants⁹, cocaine and amphetamines². Understanding the molecular basis for NSS function, together with the mechanisms of transporter modulation and inhibition by small molecules, requires determining NSS structures by atomic resolution methods such as X-ray crystallography. At present, however, NSS proteins have proven refractory to crystallization efforts.

LeuT is a prokaryotic NSS ortholog from the hyperthermophilic bacteria *Aquifex aeolicus*¹⁰ and is a valuable paradigm by which to probe relationships between atomic structure and molecular mechanism in NSS proteins. LeuT solubilized in β -OG crystallizes readily, and its structure has been solved at high resolution in the presence of substrates^{10,11}, the competitive inhibitor tryptophan¹¹ and numerous noncompetitive inhibitors^{12,13}. LeuT has also been the focus of multiple spectroscopic studies, including electron paramagnetic resonance experiments¹⁴ and single-molecule fluorescence resonance energy transfer studies^{15,16}. Furthermore, LeuT has proven a valuable template for homology modeling^{17,18}, small molecule docking^{19,20} and molecular dynamics simulations²¹ of related eukaryotic NSS proteins, as well as a general guide for design and interpretation of mutagenesis and structure-function studies of NSSs²².

Recently, however, a number of experiments have called into question the extent to which crystal structures of LeuT solubilized in β -OG reflect functionally relevant states of the transporter. Substrate binding, flux and computational studies^{23,24} have led researchers to hypothesize that β -OG interferes with the binding of a second substrate to LeuT at the S2 binding site located in the extracellular vestibule and that by binding to this site, β -OG stabilizes an inhibitor-bound conformation of the transporter²³. Furthermore, they assert that the S2 site, similarly to the S1 site, harbors a high-affinity substrate binding site and that the occupancy of the S2 site by the substrate is essential for substrate transport. By contrast, crystallographic^{10–12}, binding and substrate flux studies²⁵ are consistent with a single high-affinity substrate site. Because LeuT is the only NSS ortholog that is amenable to high-resolution structural studies, resolving these controversies is crucial to the field. Therefore, we set out to crystallize LeuT in a lipid membrane-like environment so that we could study its conformation and substrate-binding properties in the absence of β -OG and in the presence of bilayer-compatible lipids.

Here we describe multiple high-resolution crystal structures of LeuT crystallized in dimyristoyl phosphatidylcholine (DMPC) and 3-[(3-cholamidopropyl)dimethylammonio]-2-hydroxy-1-propanesulfonate (CHAPSO) bicelles, a mixture of dialkyl lipid and detergent molecules that is a more effective mimic of the native membrane bilayer^{26,27} than detergent micelles²⁸. Using LeuT protein purified in *n*-dodecyl- β -D-maltoside (C₁₂M) and never exposed to β -OG, we show that the conformations of LeuT–substrate complexes in lipid bicelles are nearly identical to the conformation of the original LeuT–leucine β -OG structure¹⁰, except for small differences in loops and crystal contact regions. Additionally, using the detergent *n*-heptyl seleno- β -D-glucoside (β -SeHG), we map the detergent sites in crystals grown

¹Vollum Institute, Oregon Health & Science University, Portland, Oregon, USA. ²Department of Biochemistry and Molecular Biology, Oregon Health & Science University, Portland, Oregon, USA. ³Howard Hughes Medical Institute, Oregon Health & Science University, Portland, Oregon, USA. Correspondence should be addressed to E.G. (gouaux@ohsu.edu).

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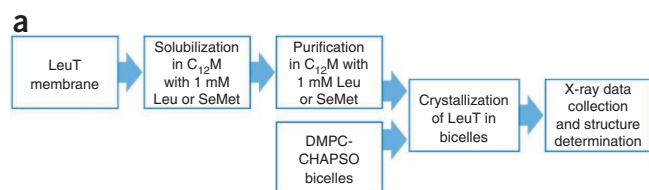


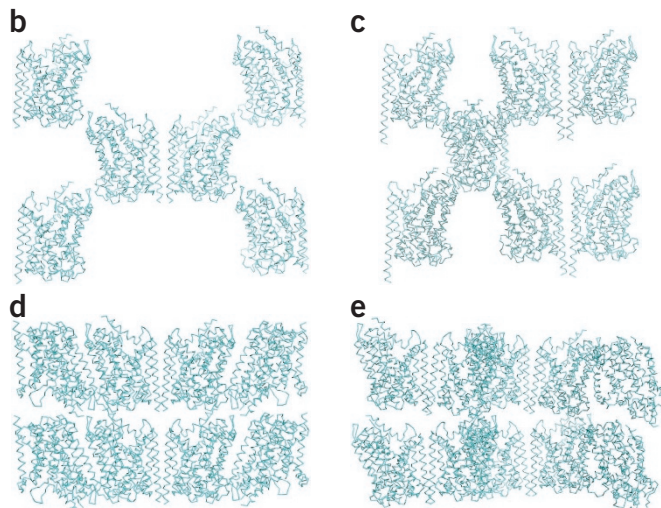
Figure 1 Experimental flow chart and lattice contacts in bicelle-based crystal forms. (a) Flow chart defining the detergents and ligands used during purification and crystallization. (b–e) Crystal packing for different crystal forms: *C*2 (LeuT–Leu) (b) and *P*2 (LeuT–Leu) (c), viewed perpendicularly to the *b* axis; *P*2₁ (LeuT–Leu, form A) (d) and *P*2₁2₁2 (LeuT–SeMet) (e), viewed perpendicularly to the *c* and *a* axes, respectively.

using micelles, and with *n*-dodecyl seleno- β -D-maltoside (C_{12} SeM), we probe the binding of detergent to bicelle-derived structures. Most notably, we find only one substrate molecule bound to the transporter, in the primary or S1 binding site. Together, these structures validate the original LeuT structure crystallized in β -OG, demonstrate that β -OG has little effect on protein conformation and strongly suggest that there is only a single high-affinity substrate site.

RESULTS

LeuT preparation and crystallization

We purified LeuT in the presence of 1 mM leucine or selenomethionine (SeMet) (Methods, Fig. 1a) in order to maximize the occupancy of substrate binding sites. We further used the detergent C_{12} M during solubilization and purification in order to minimize artifacts from



β -OG, such as its binding to the extracellular vestibule²³. Following purification, LeuT was incorporated into DMPC-CHAPSO bicelles^{26,27}, and crystals were grown by vapor diffusion at 20 °C (Fig. 1a).

The LeuT–leucine complex (LeuT–Leu) in DMPC-CHAPSO bicelles crystallized under two different conditions, yielding crystals with the space groups *C*2 and *P*2 and with diffraction limits of 2.5 Å and 3.1 Å, respectively (Table 1). To sensitively detect the presence of the substrate, we crystallized the LeuT–SeMet complex in two different crystal forms, also using DMPC-CHAPSO bicelles. These crystals belong to the space groups *C*2 and *P*2₁2₁2 and diffracted to 2.7 Å and 4.5 Å resolution, respectively (Table 2).

Because the binding of detergent molecules to LeuT is at the center of several controversies, we prepared two detergents, β -SeHG and C_{12} SeM. Crystals in the space group *C*2, which are isomorphous to the original *C*2 β -OG crystal form, were grown using LeuT that was purified and crystallized in β -SeHG. To determine whether there is ordered detergent in the bicelle-derived crystals, we purified LeuT in C_{12} SeM and grew crystals using DMPC-CHAPSO bicelles. The crystals grown from β -SeHG diffracted to 2.1 Å resolution (Table 2), whereas those derived from LeuT that were purified in C_{12} SeM and crystallized in bicelles belong to the *P*2₁ space group and diffracted to 3.5 Å and 4.5 Å resolution, respectively (Table 1).

Structure of LeuT in bicelles

All structures were solved by molecular replacement using the *C*2 β -OG (PDB 2A65)¹⁰ or the *C*2 bicelle structures as search probes (Online Methods). These initial structures were then subjected to cycles of crystallographic refinement and manual rebuilding, ultimately yielding structures with good crystallographic and stereochemical statistics. The six structures determined using crystals derived from DMPC-CHAPSO bicelles had different lattices harboring distinct protein–protein contacts, thus enabling

Table 1 Data collection and refinement statistics

	LeuT–Leu	LeuT–Leu	LeuT–Leu (<i>P</i> 2 ₁ form A)	LeuT–Leu (<i>P</i> 2 ₁ form B)
Data collection				
Wavelength (Å)	0.979	1.000	0.970	0.970
Space group	<i>C</i> 2	<i>P</i> 2	<i>P</i> 2 ₁	<i>P</i> 2 ₁
Cell dimensions				
<i>a</i> , <i>b</i> , <i>c</i> (Å)	123.8 90.8, 81.6	81.2, 84.4, 91.0	57.3, 179.9, 57.2	70.2, 112.5, 165.3
α , β , γ (°)	90.0, 103.8, 90.0	90.0, 98.0, 90.0	90.0, 89.9, 90.0	90.0, 97.6, 90.0
Resolution (Å)	40.0–2.50	40.0–3.10	50.0–3.50	20.0–4.50
	(2.59–2.50)	(3.21–3.10)	(4.14–3.50)	(4.66–4.50)
<i>R</i> _{merge}	0.08 (0.32)	0.12 (0.52)	0.14 (0.34)	0.12 (0.18)
<i>I</i> / σ <i>I</i>	10.2 (2.0)	11.2 (1.8)	6.1 (2.1)	4.6 (4.2)
Completeness (%)	97.4 (90.4)	97.0 (98.0)	94.3 (83.4)	84.0 (80.2)
Redundancy	2.6 (2.4)	3.1 (3.1)	2.7 (2.5)	2.1 (2.0)
No. reflections	29,352	21,407	13,846	12,873
Refinement				
Resolution (Å)	40.0–2.50	40.0–3.10	50.0–3.50	20.0–4.50
<i>R</i> _{work} / <i>R</i> _{free}	0.21 / 0.23	0.24 / 0.25	0.26 / 0.31	0.26 / 0.27
No. atoms				
Protein	3,968	7,936	7,936	15,872
Ligand/ion	2 Na, 1 Leu	4 Na, 2 Leu	4 Na, 2 Leu	8 Na, 4 Leu
Water	63	12	0	0
<i>B</i> -factors (Å ²)				
Protein	46.2	69.54	–	–
Ligand/ion	31.3	54.8	–	–
Water	50.4	41.5	–	–
R.m.s. deviations				
Bond lengths (Å)	0.010	0.010	0.008	0.008
Bond angles (°)	0.689	0.698	0.609	0.518

Values in parentheses correspond to the highest-resolution shells.

Table 2 Data collection and refinement statistics

	LeuT–SeMet	LeuT–SeMet	LeuT–SeMet	LeuT–β-SeHG
Data collection				
Wavelength (Å)	0.979	1.200	0.970	0.979
Space group	C2	C2	$P2_12_12$	C2
Cell dimensions				
<i>a</i> , <i>b</i> , <i>c</i> (Å)	122.3 90.1, 82.0	122.3, 90.4, 82.0	116.8, 182.7, 81.8	89.7, 86.4, 81.4
α , β , γ (°)	90.0, 103.9, 90.0	90.0, 103.6, 90.0	90.0, 90.0, 90.0	90.0, 96.2, 90.0
Resolution (Å)	40.0–2.71	40.0–3.00	40.0–4.50	50.0–2.10
	(2.81–2.71)	(3.11–3.00)	(4.66–4.50)	(2.18–2.10)
R_{merge}	0.08 (0.34)	0.09 (0.37)	0.09 (0.26)	0.08 (0.48)
$I/\sigma I$	12.2 (3.4)	14.1 (4.4)	18.4 (6.2)	21.0 (2.6)
Completeness (%)	98.8 (92.2)	99.9 (100)	94.6 (83.9)	97.4 (94.4)
Redundancy	3.1 (3.0)	4.0 (4.0)	9.1 (7.3)	5.0 (4.5)
No. reflections	23,553	17,133	10,449	35,135
Refinement				
Resolution (Å)	40.0–2.71	40.0–3.0	40.0–4.50	50.0–2.10
$R_{\text{work}}/R_{\text{free}}$	0.21/0.24	0.21/0.22	0.24/0.24	0.22/0.23
No. atoms				
Protein	3,968	3,968	7,936	3,995
Ligand/ion	2 Na, 1 SeMet, 1 I	2 Na, 1 SeMet, 1 I	4 Na, 2 SeMet	2 Na, 1 Leu, 1 Cl
Water	24	9	0	99
<i>B</i> -factors (Å ²)				
Protein	59.3	73.7	–	31.7
Ligand/ion	49.9	63.9	–	21.0
Water	55.7	65.0	–	35.9
R.m.s. deviations				
Bond lengths (Å)	0.010	0.010	0.008	0.012
Bond angles (°)	0.647	0.627	0.527	0.578

Values in parentheses correspond to the highest-resolution shells.

us to investigate the structure of LeuT in a range of crystal environments (Fig. 1b–e). A common element to all these crystal forms is a ‘parallel’ LeuT dimer, similar to that observed in the original LeuT–β-OG structure¹⁰. Matthews coefficient (V_M , the crystal volume per unit of protein molecular weight)²⁹ values for these crystal forms range from 2.6 to 3.8 Å³ Da^{−1}, giving solvent percentages from 53% to 68%. Our $P2_1$ crystal (form A) has a crystal contact interface not previously seen that involves hydrophobic interactions between TM5 helices of noncrystallographic symmetry-related molecules (Fig. 1d and Supplementary Fig. 1a,b).

Analysis of the refined LeuT–Leu complexes in both the C2 and $P2$ space groups revealed structures similar to the first LeuT structure obtained from protein crystallized in the detergent β-OG (PDB 2A65)¹⁰ (Table 3). As found previously, the LeuT protomer consists of 12 transmembrane helices (TM1–TM12), multiple solvent-exposed loops with two of the long loops—extracellular loops 2 and 4 (EL2 and EL4)—in close juxtaposition. Both structures have an occluded state conformation in which a single leucine molecule and two sodium ions occupy binding sites approximately halfway across the membrane bilayer, at sites defined mainly by TM3, TM8 and the unwound regions of TM1 and TM6. Because the C2 bicelle crystals yielded substantially higher resolution diffraction data than the $P2$ form, we use this structure for a detailed structure comparison with the C2 β-OG structure (PDB 2A65).

Comparison of LeuT structures in bicelles and in β-OG

Superimposition of the C2 bicelle and C2 β-OG crystal structures yields an overall r.m.s. deviation of 0.73 Å for main chain atoms (Fig. 2a and Table 3), thus demonstrating that the structures are highly

similar. Nevertheless, because others²³ have stated that TM6a (Pro241–Leu255) adopts a different conformation in molecular dynamics simulations from its conformation in the C2 β-OG structure owing to a β-OG molecule bound in the extracellular vestibule instead of a second substrate molecule²³, we focused our attention on this transmembrane segment. There is strong electron density for TM6a, thus allowing us to unambiguously define its position (Fig. 2b). Using the overall superposition described above, TM6a in both the C2 bicelle and C2 β-OG structures showed essentially identical conformations and positions, with an r.m.s. deviation of 0.14 Å for the Cα atoms (Fig. 2c). To test whether there is relative movement between the TM6a and ‘scaffold’ helices³⁰, we first superimposed transmembrane helices TM3, TM4, TM5, TM8 and TM12 and then compared the positions of transmembrane helices TM1, TM2, TM6, TM7 and TM11 (Supplementary Fig. 2a). Our analysis demonstrates that the conformation and position of TM6a is nearly identical for the C2 bicelle and C2 β-OG crystal structures.

Upon detailed inspection of the C2 bicelle and C2 β-OG crystal structures, however, we find two regions of structural differences in the loop and crystal contact regions. The first region is localized to the C terminus of TM11, EL6 (the turn between TM11 and TM12) and the beginning of TM12 (Val465–Val483),

where the r.m.s. deviation for the Cα atoms is 2.1 Å (Fig. 2d). In comparison to the C2 β-OG structure, the C terminus of TM11 and EL6 in the C2 bicelle structure is shifted away from TM10. In addition, residues at the C-terminal end of TM11 (Ile472–Ile475) adopt a 3_{10} -helix conformation in the C2 β-OG crystal structure, whereas in the C2 bicelle structure, this region adopts an α-helical structure. We suggest that these differences are because there are six detergent molecules near this area in the C2 β-OG structure, and because they are absent in the C2 bicelle structure, there are rearrangements in local conformation at the end of TM11 and EL6 (Supplementary Fig. 2b). Indeed, one of the β-OG molecules of the C2 β-OG crystal form makes direct contact with residues at the end of TM11 and EL6.

The C terminus of TM11 and TM12 also participate in subunit-subunit contacts in the LeuT dimer, and differences in the C2 bicelle and C2 β-OG crystal forms result in slightly different subunit-subunit contacts, such that superposition of one subunit does not result in the superposition of the second subunit (Supplementary Fig. 2b). An additional manifestation of these differences can be seen upon the inspection of Trp481, a residue at the beginning of TM12 (Supplementary Fig. 2b,c). In the C2 β-OG structure, the indole rings of symmetry-related Trp481 residues are oriented ‘tip to tip’, and the alkyl groups of β-OG molecules insert into the dimer interface. By contrast, in the C2 bicelle structure, the indole rings make extensive contacts by stacking against one another. Indeed, the differences in the contacts between Trp481 residues is probably coupled to variations in lattice contacts, and in the C2 β-OG crystal form, there are interactions between the TM11 of one molecule and TM12, IL1 and IL5 in the symmetry-related molecule (Supplementary Fig. 3a).

Table 3 Comparison of bicelle-based and C2 β -OG crystal structures

	LeuT- β -OG	LeuT- β -SeHG	LeuT-Leu (C2)	LeuT-Leu (P2)	LeuT-SeMet (C2)
LeuT- β -OG	0				
LeuT- β -SeHG	0.25	0			
LeuT-Leu (C2)	0.72	0.64	0		
LeuT-Leu (P2)	0.38	0.36	0.76	0	
LeuT-SeMet (C2)	0.73	0.64	0.06	0.76	0

Numbers in the table are r.m.s. deviation values for main chain atoms following superposition (Å).

These contacts involve electrostatic interactions between Glu470 and Arg508, Lys474 and Tyr82, and Glu477 and Lys431, interactions that are absent in the C2 bicelle crystal form.

The second region of structural difference is localized to EL2 (Val154–Lys163) (Fig. 2e). Here, the local r.m.s. deviation for the C α atoms between the C2 bicelle and C2 β -OG structures is 1.3 Å. Because these residues are involved in crystal contacts in the C2 β -OG crystal form but not in the C2 bicelle form, these conformational distinctions are straightforwardly attributable to differences in crystal contacts (Supplementary Fig. 3b).

Notably, the two regions of structural difference between the C2 bicelle and C2 β -OG crystal forms are not observed upon comparison of the P2 bicelle and C2 β -OG structures. Indeed, the P2 bicelle structure superimposes very well with the C2 β -OG structure, with an overall r.m.s. deviation of 0.38 Å for main chain atoms (Table 3). This close agreement of the two structures is undoubtedly grounded in the fact that despite the difference in space groups, the two unit cells are closely related, and the protein molecules have nearly identical lattice contacts (Supplementary Fig. 3a,b). We note, nevertheless, that the C terminus of TM11 (Ile475–Glu478) in the P2 bicelle form is disordered, thus a direct comparison of the same region in the C2 β -OG form is not possible.

Despite the overall congruence of the P2 bicelle and C2 β -OG structures, there are nevertheless two regions of difference localized to loops and crystal contacts. The first change involves the EL4 region (Val308–Phe331). In the P2 bicelle crystal structure, the sharp turn (Gly318, Ala319 and Phe320) between EL4a and EL4b in the P2 bicelle is positioned about 1 Å closer to the substrate binding site in comparison to its position in the C2 β -OG structure (r.m.s. deviation = 1.1 Å

for the C α atoms; Supplementary Fig. 3c). This difference may be attributed either to the flexibility of this loop, the details of the crystallization conditions or both. The second change is localized to the C-terminal end of TM11 at Pro473 and Lys474 (Supplementary Fig. 3c). Here the P2 bicelle structure deviates substantially from the C2 β -OG structure, as defined by an r.m.s. deviation of 2.1 Å for the C α atoms, which may be attributed to the absence of β -OG molecules around TM11.

Locations of bound detergent molecules

Multiple studies^{23,24} state that the β -OG is an inhibitor of LeuT, and by binding to the transporter, it results in the formation of an inhibitor-bound conformation that is distinctly different from a functional, substrate-bound conformation. As part of our effort to understand the interplay between the binding of β -OG to LeuT and its structure and function, we sought to unambiguously map detergent binding sites in the C2 β -OG crystal form and to determine whether, in the bicelle-derived crystal form, there are any bound detergents carried over from the purification of the transporter in C₁₂M. To do this, we synthesized selenium-containing detergents (Fig. 3) that would allow us to sensitively map detergent sites by anomalous scattering X-ray diffraction experiments.

We first determined that β -SeHG is a faithful analog of β -OG and that LeuT purified in β -SeHG forms well-diffracting crystals that are isomorphous to C2 β -OG crystals. On the basis of anomalous difference Fourier maps derived from C2 β -SeHG crystals, we unambiguously identified nine β -SeHG molecules (Fig. 3a,b), five of which were located in the same position as the β -OG molecules previously defined in the LeuT- β -OG structure. In addition, we mapped four additional β -SeHG molecules located between TM3 and TM4, between TM10 and TM11, within the extracellular vestibule and near the two-fold crystallographic axis adjacent to TM12. The first of the additional β -SeHG molecules is adjacent to a previously mapped detergent molecule that is proximal to TM4, with the head group packed against the main chain carbonyl oxygen from Leu126 and the alkyl chain enclosed by Phe167, Leu126 and Ile120. The second molecule is near TM11 and TM10. Because of the weak electron density for the alkyl chain and head group, we only mapped the position of the selenium atom. Nevertheless, based on a small conformational change within the region of Val462 to Tyr471, relative to the C2 β -OG structure (r.m.s. deviation for the C α atoms is 0.5 Å), we speculate

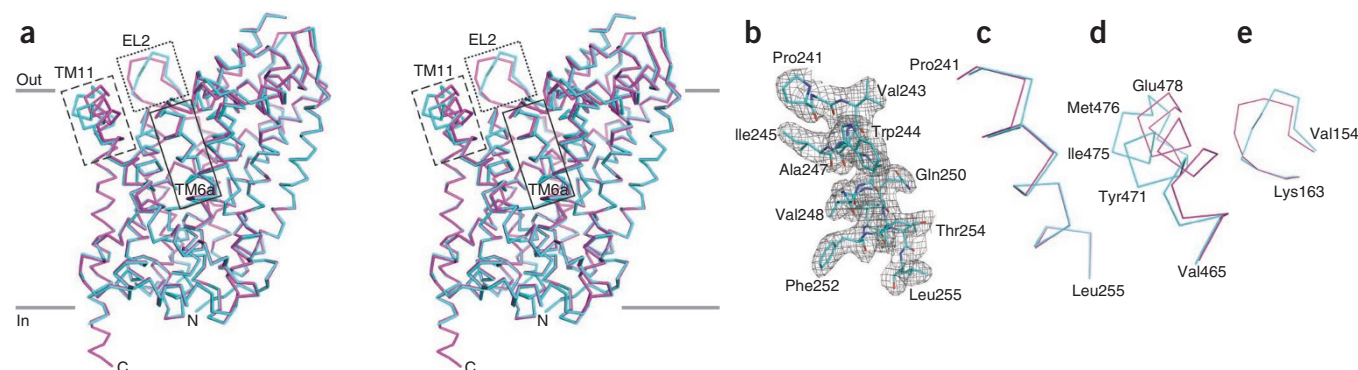


Figure 2 LeuT crystal structures derived from bicelles and β -OG are similar. (a) Stereoview of α -carbon traces of the refined LeuT structures from the C2 bicelle crystal form (cyan) and the C2 β -OG crystal form (PDB 2A65; magenta), following superposition of corresponding main chain atoms. The regions enclosed by solid, dashed and dotted lines define the TM6a (Pro241–Leu255), TM11–EL6 (Val465–Val483) and EL2 (Val154–Lys163) regions. (b) Electron density for TM6a in the C2 bicelle crystal form. Shown is electron density from a $2F_o - F_c$ map contoured at 1.5σ . Atoms in TM6a are in stick representation. (c) Closeup of TM6a, the C terminus of TM11 and EL6 (d) and the C terminus of EL2 (e). Superpositions shown in c–e are derived from those in a.

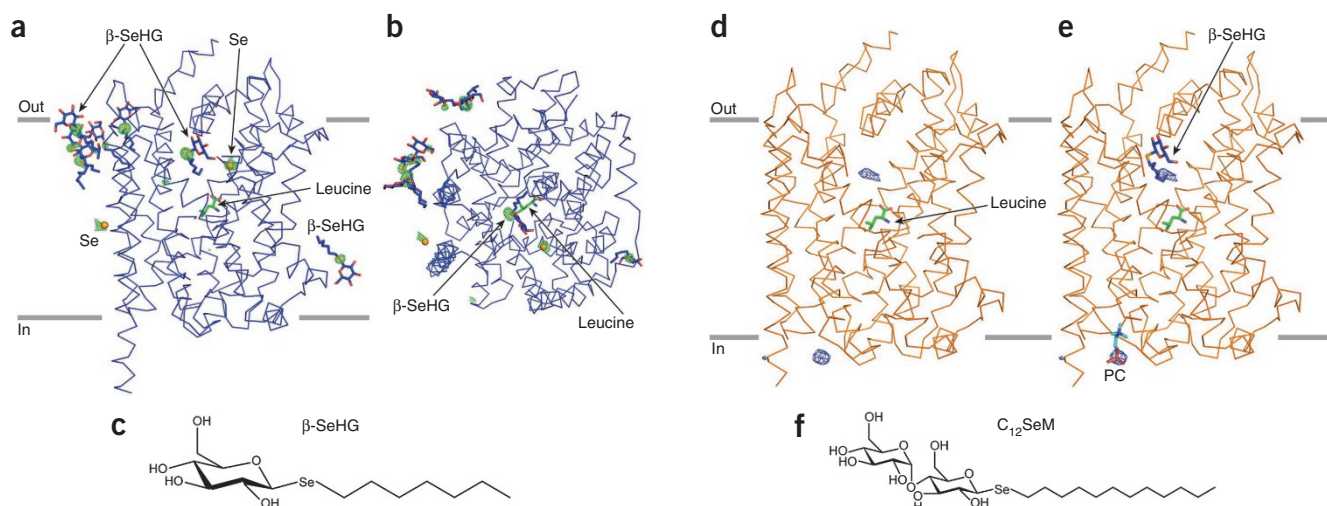


Figure 3 Mapping detergent sites in LeuT. **(a)** The anomalous difference Fourier map for LeuT (blue) crystallized in β -SeHG. The map is contoured at 5σ , is depicted in green mesh and unambiguously shows the positions of bound detergent molecules. Seven β -SeHG molecules are modeled and shown in stick representation. The positions of two additional β -SeHG molecules are inferred on the basis of strong anomalous difference peaks. Here we have only modeled the selenium atoms (orange spheres), owing to weak electron density for the headgroups and alkyl chains. Leucine in the primary site is shown in stick representation (green), with several residues omitted for clarity. **(b)** The anomalous difference map for LeuT- β -SeHG viewed from the extracellular side of the protein. **(c)** The chemical structure of β -SeHG. **(d)** The $F_0 - F_c$ difference electron density map for LeuT-Leu ($P2_1$ form A, chain A, orange) crystallized in bicelles. The protein was purified using C_{12} SeM. This map is shown at 4.5σ , and is depicted in blue mesh. **(e)** The two peaks in **d** are in similar positions to those for the alkyl chain of β -SeHG in the LeuT- β -SeHG structure and the phosphocholine molecule (PC) in the LeuT-Leu ($C2$ form) structure, respectively. Both β -SeHG and phosphocholine molecules are shown as stick representations. **(f)** The chemical structure of C_{12} SeM.

that the aliphatic chain forms a direct hydrophobic interaction with the indole ring of Trp467. This speculation is also consistent with the observation arising from the LeuT-Trp structure (PDB 3F3A)¹¹, in which the alkyl chain of a β -OG molecule in a similar position is enclosed by Ile410, Val413, Leu463 and Trp467. The third β -SeHG molecule is located in the extracellular vestibule, where the alkyl chain of β -SeHG is in a hydrophobic pocket formed by the side chains of Leu25, Leu29, Ile111, Phe253, Phe320 and Leu400. This molecule shares a very similar position to a previously determined β -OG molecule (PDB 3GJD)²³ and has been proposed to act as an inhibitor of LeuT, supposedly displacing a substrate molecule from the S2 binding site and distorting the conformation of TM6a (ref. 23). The last detergent is located close to the two-fold axis of TM12. Again, we only modeled the selenium atom because of the weak electron density of the head group and the alkyl chain.

Unfortunately, in the first bicelle-derived LeuT-Leu crystals ($P2_1$ form A) prepared using protein purified in C_{12} SeM, we were not able to measure the anomalous signal of C_{12} SeM because of the twin properties of this crystal form (about 40% twin fraction). However, from $F_0 - F_c$ electron density maps, we found two prominent peaks located around IL1 and the extracellular vestibule, respectively (Fig. 3d). After superimposing this $P2_1$ bicelle structure with the high-resolution C2 bicelle and LeuT- β -SeHG structures, we found the peak around IL1 is in a position similar to that of the phosphocholine molecule modeled in the C2 bicelle structure, whereas the other peak in the vestibule is close to the alkyl chain of β -SeHG in the LeuT- β -SeHG structure (Fig. 3e). Based on these parallels, we suggest that the peak around IL1 corresponds to a phosphocholine molecule, and the peak in the vestibule corresponds to the alkyl chain of a lipid molecule. Because we were able to detect the phosphocholine group in an $F_0 - F_c$ map, we should be able to detect selenium in C_{12} SeM in the same difference maps. Thus, because there are no other prominent peaks in the $F_0 - F_c$ electron density maps resulting from bicelle-derived crystals for which LeuT was prepared

in C_{12} SeM, we suggest that there are no tightly bound detergent molecules in these LeuT crystals from bicelles.

In addition, we crystallized another type of LeuT-Leu crystal ($P2_1$ form B) that was prepared using protein purified in C_{12} SeM, which had different cell dimensions from the previous $P2_1$ form A structure (Table 1). More notably, we were able to measure the anomalous signal of C_{12} SeM, because there is no twin fraction in this crystal form. Not unexpectedly, we found no substantial peaks in anomalous difference Fourier maps that were calculated using diffraction data measured at the selenium K absorption peak (Supplementary Fig. 4), thus further demonstrating that there are no tightly bound detergent molecules. These results suggest that unlike β -OG, C_{12} M does not occupy the extracellular vestibule and that it is unlikely that the transporter structure in bicelles is perturbed by the presence of bound C_{12} M detergent molecules.

Determination of substrate site location and stoichiometry

The number of high-affinity substrate binding sites in the occluded-state form of LeuT is a highly contentious topic. On the one hand, some^{15,23,24} argue that there are two high-affinity sites. On the other hand, experiments from another group^{10–12,25} are entirely consistent with a single high-affinity site. The first group states that β -OG displaces a substrate molecule from the secondary or S2 site and that is why, in the C2 β -OG crystal structures, no substrate is observed in the S2 site. Because we defined the conditions under which to grow well-ordered LeuT crystals in the absence of β -OG and in the presence of bilayer-like bicelles, we decided to examine whether we could locate a second substrate molecule in the S2 site by X-ray diffraction methods. To do this, we carried out two different experiments.

We first carefully examined the electron density in the C2 bicelle crystals of the LeuT-Leu complex in which 1 mM leucine was present throughout purification, crystallization and cryoprotection. In doing this, we clearly identified one leucine molecule and two sodium ions in the primary binding pocket, on the basis of unbiased $F_0 - F_c$ omit

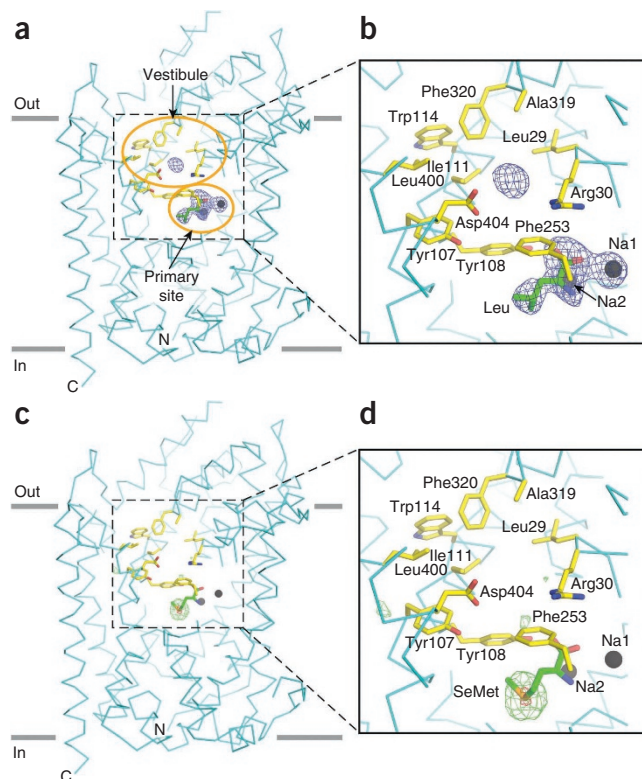
Figure 4 LeuT has a single high-affinity substrate site in bicelles.

(a) Electron density map for the primary substrate binding site and extracellular vestibule in the DMPC-CHAPSO bicelle C2 crystal form of the LeuT–Leu complex. The substrate leucine in the primary site (green) is in stick representation, and two sodium ions are shown as black spheres. Critical residues in the vestibule are in stick representation. The $F_o - F_c$ difference electron density map is shown at 3σ , is depicted in blue mesh and is calculated with the substrate leucine and the two sodium ions omitted (leucine and sodium are shown for reference). (b) Closeup of primary binding site and the vestibule. (c) Anomalous difference Fourier map for primary substrate site and extracellular vestibule derived from diffraction data measured from LeuT–SeMet (C2 form) crystals. The map is contoured at 5σ and 20σ and is depicted in green and red mesh, respectively. SeMet was positioned in the primary site and is shown in stick representation. Two sodium ions (Na1 and Na2) are illustrated as black spheres (d) Closeup of primary binding site and the vestibule.

electron density maps (Fig. 4a,b), which were exactly like those seen in the original C2 β -OG crystal structure¹⁰. If the putative S2 binding site is also a high-affinity substrate site, then we should find similar electron density in the S2 site, near Ile111 and Leu400 (refs. 23,24). However, we could not find a similar electron density in the putative S2 site (Fig. 4a,b). Instead, we found only a small, egg-shaped electron density feature in the extracellular vestibule, approximately 4.7 Å from Ile111, 5.8 Å from Arg30 and 6.8 Å from Leu400. We assert that this density is not attributable to a bound leucine molecule for two reasons. First, the shape of the density is not well fit by a leucine molecule. Second, the density is not strong enough to be a well-ordered leucine molecule, and if we do model a leucine molecule in this position, the resulting B -factor of 111 Å² is more than twice that of the mean B -factor of 46.2 Å². Although it is difficult to conclusively define molecular identity at 2.5 Å resolution, we modeled different entities into the density and compared the refined B -factors. When we fit a single water or a Cl ion to the density, the resulting B -factors were 32 Å² and 61 Å², respectively. However, analysis of anomalous difference maps measured from crystals grown in the presence of iodide did not yield any prominent peaks near this electron density feature (Supplementary Fig. 5a), and thus we suggest that this density is due to the presence of several water molecules or the alkyl chain from a lipid molecule.

We next purified LeuT and grew crystals in the presence of the substrate analog SeMet, knowing that if there was any substantial occupancy of the S2 site by a substrate molecule, we would easily pick it up in anomalous difference Fourier maps. We obtained two different bicelle-based crystal forms of the LeuT–SeMet complex, one of which strongly diffracted X-rays beyond 2.7 Å resolution. Most notably, after careful analysis of maps from both crystal forms, we found that there is a strong anomalous difference electron density in the primary or S1 site (24 σ) but no substantial anomalous difference electron density in the putative S2 site or anywhere else in the transporter (Fig. 4c,d and Supplementary Fig. 5b,c).

To verify that SeMet is a bona fide substrate of LeuT, we would ideally carry out substrate flux experiments using ³H-SeMet. This isotope, however, is unavailable. Therefore we asked whether SeMet inhibits LeuT-catalyzed ³H-Ala flux to the same extent as does methionine, a known substrate¹¹. From these experiments, we found that 1 mM SeMet inhibits ³H-Ala uptake by LeuT to a similar extent as does 1 mM methionine (Supplementary Fig. 6), consistent with our hypothesis that SeMet is a faithful analog of methionine and that SeMet should therefore label the substrate binding sites. Taken together, these diffraction experiments using bicelle-derived crystals are consistent with the conclusion that there is a single high-affinity substrate site in LeuT.



DISCUSSION

The LeuT–substrate structures observed from multiple crystal forms in DMPC-CHAPSO bicelles provide excellent molecular blueprints by which to explore fundamental molecular principles of structure, mechanism and pharmacology in NSS family members. The marked similarity between the structures in lipidic environments and detergents suggests that the overall conformation of the substrate-bound, occluded state of LeuT is mainly independent of crystal environment and condition, with exceptions made for the local conformations of a few surface-exposed loops, including EL2, EL4 and EL6. Indeed, these three loops are probably dynamic in nature, and in the context of NSS proteins, have been implicated in conformational dynamics associated with substrate translocation, on the basis of biochemistry and spectroscopic studies^{14,31–33}.

Our results further demonstrate that the original LeuT structure crystallized in β -OG micelles represents a functional conformational state of the transporter. We suggest that this is the case, based on detailed comparisons of the bicelle and β -OG structures of LeuT, for which we find that the β -OG molecules do not substantially affect the overall conformation of LeuT. The most substantial differences between the β -OG and bicelle-based crystal structures are localized to loops and are probably the consequence of differences in the detergent-lipid environment and the lattice contacts. Indeed, the conformational differences that we detect in TM11 and EL6 upon comparing the LeuT– β -OG with our bicelle-derived LeuT–Leu structures prompt us to compare our analysis to a recent EPR analysis demonstrating that the distance between E478C in TM11 and E236C in EL3 becomes closer when β -OG is added to LeuT in the presence of leucine and sodium ions¹⁴. Although this relative distance change was interpreted as a movement of TM6a¹⁴, our analysis suggests that the bound β -OG molecule in the extracellular vestibule does not affect the conformation of TM6a and that the salient conformational change is rooted in the movements

of TM11 and EL6 that are due to their interactions with β -OG molecules on the periphery of the protein.

The fact that we could not observe a substrate in the proposed S2 site^{23,24} of these bicelle-derived structures demonstrates that even when LeuT is purified in $C_{12}M$ and crystallized in a membrane-like environment, we are unable to visualize a substrate in the S2 site. A confounding issue in resolving differences between structural studies and functional studies is that the β -OG detergent has been regularly used in crystallization experiments^{10–12}, whereas $C_{12}M$ has been used in binding assays^{23–25}. Indeed, a β -OG molecule was identified in the extracellular vestibule of LeuT in a previous study²³ and also in the work reported here. Most notably, the previous study suggests that the presence of β -OG interferes with measurement of stoichiometry in LeuT²³. Because we entirely avoided β -OG in these bicelle-based crystal forms, the crystallographic studies reported here allow visualization of the substrate sites of LeuT in the absence of β -OG, and thus demonstrate that β -OG cannot account for the differences that the two other groups observed in the substrate binding stoichiometry of LeuT^{10,11,15,23–25}. We further note that tricyclic antidepressants with micromolar affinity to the S2 site compete with the β -OG molecule bound to the vestibule, readily occupying the S2 site in the crystal¹². These observations raise the question of why a substrate, with nanomolar affinity²³, cannot similarly bind to the S2 site. Although we found nonprotein electron density in the extracellular vestibule of the bicelle-derived LeuT structures, we suggest that it represents the alkyl chain of a lipid molecule or several ordered water molecules.

SeMet was used to further investigate whether the nonprotein electron density in the extracellular vestibule could be attributed to a partially occupied substrate. Extensive prior work has shown that SeMet is a good substitute for methionine, as judged by the similar activity and properties of selenomethionyl and natural proteins³⁴, by the similarity of SeMet and methionine determined from comparisons of their high-resolution crystal structures³⁵, and by the similarity of SeMet and methionine functions in biosynthesis³⁶. Based on the fact that SeMet is generally a viable analog of methionine and that it inhibits ³H-Ala uptake similarly to methionine, we used it to probe the substrate binding sites in LeuT.

The resulting anomalous difference electron density maps from LeuT–SeMet structures demonstrate that the nonprotein density in the extracellular vestibule is unlikely to be attributable to a substrate molecule and that there is a single SeMet molecule bound to the S1 site. Whereas the different stoichiometry of substrate binding to LeuT determined from functional studies^{23–25} might be due to different biochemical methodologies³⁷, the structural data presented here show that LeuT crystallized in lipid bicelles contains a single high-affinity substrate binding site. Nevertheless, our data do not rule out the possibility that the proposed S2 site may be a transiently occupied, low-affinity binding site for the substrate as it moves from the extracellular vestibule to the S1 site^{11,38}.

In conclusion, the multiple LeuT structures in complex with substrate and substrate analog in lipid bicelles are consistent with the original LeuT– β -OG structure. Because β -OG has only minimal effects on LeuT conformation, our analysis demonstrates that the LeuT structures crystallized in β -OG are functional forms of the transporter and in all likelihood represent functionally relevant states along the transport cycle of LeuT.

METHODS

Methods and any associated references are available in the online version of the paper at <http://www.nature.com/nsmb/>.

Accession codes. Coordinates and structure factors for LeuT–Leu ($C2$ bicelle), LeuT–Leu ($P2$ bicelle), LeuT–Leu ($P2_1$ form A bicelle), LeuT–Leu ($P2_1$ form B bicelle), LeuT–SeMet ($C2$ bicelle), LeuT–SeMet ($C2$ bicelle, collected at 1.2 Å), LeuT–SeMet ($P2_12_12$ bicelle) and LeuT– β -SeHG ($C2$ β -SeHG) have been deposited in the Protein Data Bank with codes 3USG, 3USI, 3USJ, 3USK, 3USL, 3USM, 3USO and 3USP, respectively.

Note: Supplementary information is available on the Nature Structural & Molecular Biology website.

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AUTHOR CONTRIBUTIONS

H.W. and E.G. designed the research; H.W., J.E. and E.G. carried out the research and analyzed the data; and H.W. and E.G. wrote the paper.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Protein expression, purification and crystallization. LeuT was expressed and purified as described previously¹⁰, with two exceptions. In purification of the LeuT–Leu complex, 1 mM L-leucine was included in all solutions, and C₁₂M was the only detergent used throughout the purification. After cleavage of the octahistidine tag by thrombin, the LeuT–Leu complex was further purified by size-exclusion chromatography (SEC) in 20 mM Tris-HCl, pH 8.0, 50 mM NaCl, 1 mM leucine and 0.7 mM C₁₂M. Bicelles were prepared by mixing DMPC and CHAPSO at a molar ratio of 2.8:1 with water to yield a final concentration of 35% (w/v)²⁷. LeuT–Leu (8–10 mg ml^{−1}) in the presence of 7% (w/v) DMPC-CHAPSO bicelles was crystallized by vapor diffusion at 20 °C, with the crystallization reservoir solution containing 100 mM sodium acetate, pH 4.5–5.0, 25–35% (v/v) 2-methyl-2,4-pentanediol (MPD) and 5–10% (v/v) PEG 400. For LeuT in C₁₂SeM (Anatrace), 0.7 mM C₁₂SeM was substituted for C₁₂M in the final size-exclusion chromatography stage. Crystals were directly flash frozen in liquid nitrogen before collection of X-ray diffraction data.

The LeuT–SeMet complex was purified as described for the LeuT–Leu complex, except that 0.1–1 mM SeMet was included in all solutions. In the final size-exclusion chromatography stage, we used 20 mM Tris-HCl, pH 8.0, 50 mM NaCl or NaI, 1 mM SeMet and 0.7 mM C₁₂M. Bicelles used for crystallization of the LeuT–SeMet complex were prepared by replacing 10% (w/v) of DMPC with DMPE. For growth of the LeuT–SeMet crystals (P₂₁2₁2 form), there was 50 mM NaCl in the SEC buffer, whereas for the LeuT–SeMet crystals (C2 form), NaCl in the SEC buffer was replaced with 50 mM NaI. LeuT–SeMet crystals (P₂₁2₁2 form) were grown in the presence of 7% (w/v) DMPC-DMPE-CHAPSO bicelles at 20 °C in crystallization buffer with 100 mM sodium acetate, pH 5.1, 32% (v/v) MPD, 8.5% (v/v) PEG 400, 50 mM MgCl₂ and 100 mM urea. LeuT–SeMet crystals (C2 form) were grown in crystallization buffer with 100 mM sodium acetate, pH 4.7, 35% (v/v) MPD, 10% (v/v) PEG 400 and 50 mM MgCl₂.

LeuT–β-SeHG was purified and crystallized as described previously¹⁰, except that β-SeHG (Anatrace) was substituted for β-OG in the final size-exclusion chromatography stage.

Data collection and structure elucidation. Diffraction datasets were collected at the Advanced Light Source (beamlines 5.0.2 and 8.2.1) and the Advanced Photon Source (beamlines 24-ID-E and 24-ID-C). Data sets were indexed, integrated and scaled using HKL2000 (ref. 39).

The LeuT–Leu structures (C2 and P2 forms) were determined by molecular replacement with Phaser⁴⁰, using the original LeuT–β-OG structure (PDB 2A65) as a search probe. After an initial solution was found, Phenix⁴¹ was used to generate the simulated annealing composite omit electron density maps. Following manual adjustment in Coot⁴² into the composite omit maps, refinement of the model against the X-ray data using Phenix was carried out until satisfactory

model statistics were obtained. Regions with weak electron density were excluded in the final model. The final C2 bicelle LeuT–Leu model consists of residues 5–131 and 135–509 of LeuT, one leucine, two sodium ions, one phosphocholine headgroup, one decane molecule, one acetate, one polyethylene glycol and 63 water molecules.

The LeuT–SeMet (C2 form and P₂₁2₁2 form) and LeuT–Leu (two P₂₁ forms) structures were determined by molecular replacement using the LeuT–Leu (C2 form) above as a search probe. The LeuT–Leu (P₂₁ forms) and LeuT–SeMet (P₂₁2₁2) structures are low-resolution structures and only rigid-body, NCS and TLS refinements were carried out.

The structure quality analysis was carried out using Molprobit⁴³. For all structures, the Ramachandran geometry is excellent, with at least 97% of the residues in the most favored regions and none in disallowed regions. All structure figures were generated with PyMOL (DeLano Scientific).

Anomalous map calculation. For the LeuT–SeMet (C2 and P₂₁2₁2 forms), LeuT–Leu (P₂₁ form B) and LeuT–β-SeHG crystallographic analyses, the anomalous difference maps were calculated using the fast Fourier transform in the CCP4 suite⁴⁴. The anomalous difference electron density map for P₂₁2₁2 LeuT–SeMet was then analyzed by real space averaging between the two subunits in the asymmetric cell, using the “NCS maps” tool of Coot⁴².

Inhibition of transport assay. LeuT was reconstituted into lipid vesicles as described before^{10–12}. Transport was carried out at 27 °C by diluting LeuT proteoliposomes by a factor of 60 into external buffer (20 mM HEPES-Tris, pH 7.0, 100 mM NaCl) containing 50 nM ³H-Ala (83 Ci mmol^{−1}) alone, with 1 mM L-methionine, and with 1 mM L-SeMet, respectively. Reactions were terminated after 5 min by transferring 150-μL aliquots of each reaction mixture into 1.5 ml cold internal buffer (20 mM HEPES-Tris, pH 7.0, 100 mM KCl). Reaction mixtures were filtered and analyzed as previously described^{10–12}. Nonspecific uptake was determined by conducting the experiment using liposomes devoid of protein. The entire experiment was conducted twice and in triplicate each time.

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